

Strategic Case Study Examination

5-Step Answering Approach – Demo and Debrief of Sample Exam Task

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战略级案例分析考试

五步答题法介绍- 样题分析

2024年11月/2025年2月考季

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The 5-step answering approach (for each exam task)

1. Read the task to identify sub-sections

- Each section will contain 2, 3 or 4 requirements of unequal point value with suggested time allocation as % of hour/ 33 marks
- Single sentence = 1 requirement
- Single sentence containing 'and' = 2 requirements

2. Type tasks and key words as bold headings in your document

- Marks are allocated under headings
- Separate key terms to help generate thoughts ('investors' and 'staff' not 'investors and staff')

3. Read the reference materials

- Relate information to the tasks and headings
- Identify any particular emphasis
- Identify how information affects task

4. Type your key points under the headings

- Keep it to a few simple words
- Try to find 5 things to say under each heading

5. Expand your answer for the time available

- Make points into short sentences.
- Relate to the pre-seen/unseen and give examples)
- Use 'what' – 'why' (... 'because' ...) approach wherever you can.
- Think/write as a 'Senior Finance Manager' – advice the Board.

5步答题法 (针对每个问题)

1. 审题并明确有多少道小题
2. 把问题以及关键词作为答题的标题，打在答题框里
3. 阅读参考资料
4. 在标题下输入你的要点
5. 在时间允许的情况下拓展你的回答

- 一个题目可能包含2, 3 或者4个小题要求，根据总分33分以及小题要求个数来分配相应的答题时间比例
- -1个句子=1个要求
- -包含“以及”的一个句子=2个要求

- -按每个标题下的内容给分
- -区分关键术语有助于发散思维，例如（“投资者”和“员工”，而不是“投资者和员工”）

- -标记出与题目和标题有关的信息
- -确认重点
- -确认信息如何影响题目

- -简短扼要
- -每个标题下尝试找出五个要点

- 将要点转化为简短的句子。
- 关联预览材料并给出例子。
- 尽可能使用“什么”-“为什么”（……“因为”……）的方法。
- 以“高级财务经理”的身份思考/回答——向董事会提供建议。

Given the above, I need your input as follows:

- Could you explain the types of cyber risks that we might face, and how they might be mitigated?

55% of 33 marks = 18 marks

So, 9 marks for each part of this sub-task.

[Sub task (1a) - 55%]

- And could you evaluate the different debt finance methods that could be used to fund a digital strategy project, and the impact of the extra debt funding required for the digital strategy project?

[Sub task (1b) - 45%]

The complete Task and Unseen Material is provided separately.

鉴于此，我需要你就以下几点提供意见：

- 你能解释我们可能面临的网络风险类型以及如何缓解这些风险吗？

33分的55%等于18分

因此，这个小题的每个部分为9分。

【小题（1a） - 55%】

- 你能评估用于资助数字战略项目的不同债务融资方法，及额外债务融资对该项目的影响吗？

【小题（1b） - 45%】

完整的题目和未事先提供的材料将分开提供。

Task 1 (a): Marking Matrix

	No rewardable material	0
Level 1	Little knowledge of cyber risk types shown and applied	1-3
Level 2	Some knowledge of cyber risk types shown and applied, and relates to the scenario.	4-6
Level 3	Good knowledge and understanding shown of cyber risk types, related to the scenario and references the pre-seen.	7-9

第一题1 (a): 评分矩阵

	无可评分内容 0分
第一级	展示并应用了很少的网络风险类型知识 1-3分
第二级	展示并应用了一定的网络风险类型知识，且与场景相关 4-6分
第三级	展示了良好的网络风险类型知识和理解，能够与场景相关联，并引用了案例预览材料 7-9分

Task 1 (a): Types of cyber risks – Key Points

Types of cyber risks at Rotomyne

Hackers

Social engineering

Malware attacks

Ransomware

Trojans

题目 1 (a): 网络风险的类型 – 要点

Saefwell的网络风险类型

黑客

社会工程学

恶意软件攻击

勒索软件

木马程序

Task 1 (a): Types of cyber risks

[Point #1] Hackers

(What is hacking?) Hacking is when authorised or unauthorised access is gained to a computer system, generally via the Internet. The person who does this is called a hacker. Authorised hackers (also known as White Hat or ethical hackers) may be employed to test the company's systems for vulnerabilities.

An unauthorised hacker (also known as Black Hat hackers, or commonly just 'hackers') is malicious and violates security to either steal, delete or modify data. They use techniques such as social engineering to access systems to install malware or carry out application attacks (I have explained below).

题目 1 (a): 网络风险类型

[要点 #1] 黑客

(什么是黑客?) 黑客是指通过互联网非法或合法访问计算机系统的行为，进行这一操作的人被称为黑客。被授权的黑客（也称为白帽黑客或道德黑客）可能被雇用来测试公司系统的安全漏洞。

未经授权的黑客（也被称为黑帽黑客，通常简称为“黑客”）带有恶意意图，违反安全规定以窃取、删除或更改数据。他们使用诸如社会工程等技术访问系统，安装恶意软件或执行应用攻击（下文将讨论）。

Task 1 (a): Types of cyber risks

(Why is hacking a cyber risk?) Unauthorised hackers present significant cyber-risks to organisations like Rotomyne, as they can target the organisation and tailor their approach to maximise the chance of succeeding in whatever malicious actions they intend. This is in comparison to other more general threats, such as computer viruses, which may be less effective in breaching Rotomyne's standard cybersecurity defences.

题目 1 (a): 网络风险类型

(为什么黑客是网络风险?).这样的黑客对像罗托矿业这样的组织构成重大的网络风险,因为他们可以专门针对该组织,并制定策略,以最大化其恶意行动的成功率。与其他更通用的威胁,如计算机病毒相比,这些威胁在突破罗托矿业标准的网络安全防御时可能效果较差

Point #1

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要点 #1

黑客

(什么是黑

客?) 黑客是指通过互联网非法或合法访问计算机系统的行为，进行这一操作的人被称为黑客。被授权的黑客（也称为白帽黑客或道德黑客）可能被雇用来测试公司系统的安全漏洞。

未经授权的黑客（也被称为黑帽黑客，通常简称为“黑客”）带有恶意意图，违反安全规定以窃取、删除或更改数据。他们使用诸如社会工程等技术访问系统，安装恶意软件或执行应用攻击（下文将讨论）。

(为什么黑客是网络风险?) 这样的黑客对像罗托矿业这样的组织构成重大的网络风险，因为他们可以专门针对该组织，并制定策略，以最大化其恶意行动的成功率。与其他更通用的威胁，如计算机病毒相比，这些威胁在突破罗托矿业标准的网络安全防御时可能效果较差

Task 1 (a): Types of cyber risks

[Point #2] Social engineering

(What is social engineering?) Social engineering refers to where malicious actors manipulate users into carrying out activities or disclosing confidential information (e.g. their login data). Methods used can include phishing attacks (a generic email sent to lots of people in order to trick recipients into downloading malware) or spear phishing (where the phishing targets a specific user, e.g. the CFO).

题目 1 (a): 网络风险类型

[要点#2] 社交工程

(什么是社交工程?) 社交工程是指恶意行为者通过操纵用户来执行特定活动或泄露机密信息（例如，他们的登录数据）。常用的方法包括钓鱼攻击（向大量人发送一般性电子邮件，目的是诱使收件人下载恶意软件）或针对性钓鱼（钓鱼攻击专门针对特定用户，比如首席财务官）。

Task 1 (a): Types of cyber risks

(Why is social engineering a cyber risk?) This is an area that is typically very difficult to manage as it plays on human psychology to achieve the desired result. However, more so than many organisations in different industries, Rotomyne is yet to become very security-conscious and incorporate it into our corporate culture. On the other hand, due to the sensitive nature of the information we gather about our customers and their activities (e.g., production schedule of battery manufacturing, car manufacturing etc.), we may be targeted, and more sophisticated social engineering methods could be employed.

题目 1 (a): 网络风险类型

(为什么社交工程是网络风险?) 这是一个通常很难管理的领域，因为它利用人的心理特点来达到预期效果。然而，与许多其他行业的组织相比，罗托矿业无疑非常注重安全。但另一方面，由于我们搜集的有关客户和业务活动的信息性质敏感，我们可能成为攻击目标，可能会遭遇更复杂的社交工程攻击。

Point #2

Social engineering

Social engineering refers to where malicious actors manipulate users into carrying out activities or disclosing confidential information (e.g. their login data). Methods used can include phishing attacks (a generic email sent to lots of people in order to trick recipients into downloading malware) or spear phishing (where the phishing targets a specific user, e.g. the CFO).

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要点 #2

社会工程

社会工程指的是恶意行为者通过操纵用户来促使其执行某些操作或泄露机密信息（例如登录凭证）。常用手段包括网络钓鱼攻击（即向大量人群发送伪装邮件，诱使他们下载恶意软件）或鱼叉式网络钓鱼（针对特定用户的钓鱼攻击，例如首席财务官）。

这一领域通常很难应对，因为它利用了人类的心理弱点来实现目的。然而，罗托矿业相比许多其他行业的组织，还没有充分重视安全问题，也未将其融入到企业文化中。另一方面，考虑到我们收集的关于客户及其业务活动的敏感信息（如电池生产计划、汽车制造等），我们可能成为攻击目标，且对方可能会使用更复杂的社会工程手段。

Task 1 (a): Types of cyber risks

[Point #3] Malware attacks

(What are malware attacks?) Malware is a broad term that includes any form of malicious software programme or code created by a hacker.

(Why are malware attacks a cyber risk?) These programmes can perform a number of tasks, such as stealing, changing or encrypting data. The motivation for creating and using these programmes varies but it is most often to cause downtime to negatively impact the business or to extort money. There are many common types of malware, including the following.

Computer viruses and worms spread from host to host in order to replicate and are designed to negatively change the way a computer or legitimate programme works. The distinction is that worms can activate themselves, whereas a virus must be activated by the user's actions. For instance, when the programme with the virus is executed, the virus might delete all of the files on the computer's hard drive.

题目 1 (a): 网络风险类型

[要点#3] 恶意软件

(什么是恶意软件?) 恶意软件是一个涵盖了由黑客创建的各种恶意软件程序或代码的广泛术语。

(为什么恶意软件是网络风险?) 这些程序可以执行许多任务，如盗取、更改或加密数据。创建和使用这些程序的动机多种多样，但通常是为了造成业务中断，对企业产生负面影响，或是进行勒索。

常见的恶意软件类型包括以下几种：

计算机病毒和蠕虫能在宿主之间传播以复制自身，并被设计为负面改变计算机或合法程序的工作方式。区别在于，蠕虫可以自动激活，而病毒则需要用户的操作来激活。例如，当执行含有病毒的程序时，病毒可能会删除计算机硬盘上的所有文件。

Task 1 (a): Types of cyber risks

[Point #4] (What is ransomware?) Ransomware is a programme that prevents us from accessing our data, information or the whole computer system until we pay a set amount of 'ransom' money. *(Why is ransomware a cyber risk?)* This is an increasingly common threat for businesses, as it is often an effective way for hackers to extort money from companies.

[Point #5] (What are trojans and why they are cyber risks?) Other common forms of malware are trojans, which pretends to be useful but will actually perform malicious functions once installed. They look an innocent program but once installed, acts to destroy the system or to provide a 'backdoor' to allow hackers to enter the system.

题目 1 (a): 网络风险类型

[要点 #4] (什么是勒索软件?) 勒索软件是一种程序，它阻止我们访问自己的数据、信息或整个计算机系统，直到我们支付一定金额的“赎金”。**(为什么勒索软件是网络风险?)** 这已成为企业面临的一种日益常见的威胁，因为这是黑客从公司敲诈钱财的一种常见方式。

[要点 #5] (什么是恶意软件以及为什么是网络风险?) 其他常见的恶意软件形式包括特洛伊木马，这种软件表面上似乎有用，但一旦安装，实际上会执行恶意功能；间谍软件，能够记录被感染系统上的活动并向黑客报告；应用程序攻击，这种攻击专门针对应用程序，因为它们在计算机和智能手机上越来越流行，而且防御难度较大。

Task 1 (a): Important tips!

Role of the Senior Finance Manager – **advise the Board.**

Identify what the Financial Director (or the Board) needs to know (which may go beyond what they actually asked for).

So, this can be the concluding paragraph.

Like with social engineering, malware is comparably less likely to be a major threat to a company like Rotomyne as we are security conscious and have defences in place, which will be described further down. However, we must not become complacent as malware is always evolving and techniques for infecting systems are likewise becoming more complex and effective, so even more secure companies can be susceptible.

题目 1 (a): 重要技巧!

高级财务经理的角色 - 向董事会提供建议。

确定财务总监（或董事会）需要了解的信息（这可能超出他们实际提出的要求）。

因此，这可以是结论段落。

与社交工程相比，对于像罗托矿业这样注重安全的公司来说，恶意软件相对不太可能构成主要威胁，因为我们已经部署了防御措施。然而，我们不能放松警惕，因为恶意软件始终在不断进化，侵入系统的技术也日益复杂和有效，因此即使是更安全的公司也可能遭到攻击。

Task 1 (a): Check list

Have we answered the question (**set by the examiner!**)?

”Could you explain the types of cyber risks that **we** might face?”

Have you related your answer to the pre-seen and unseen material?

Did you think and write as the Senior Finance Manager of Rotomyne?

第1 题(a): 检查清单

我们回答了问题了吗（由考官提出的！）？

“你能解释一下**我们**可能面临的网络风险类型吗？”

你是否将你的答案与预览材料和未提供的材料联系起来？

你是否以罗托矿业的高级财务经理的身份思考并撰写？

Task 1 (b): Mitigation of cyber risks – Key Points

Mitigation of cyber risks at Rotomyne

Protection: User security controls

Protection: Data security controls

Protection: Software security controls

Protection: Hardware security controls

Protection: Network security controls

Detection

Response

题目 1 (b): 网络风险缓解-要点

Saefwell的网络风险缓解

防护级别：保护、检测和控制

保护 - 用户安全控制

保护 - 数据安全控制

保护 - 软件安全控制

保护 - 硬件安全控制

保护 - 网络安全控制

检测

响应

Task 1 (a): Mitigation of cyber risks

Level 1	Few controls identified and few or no reasons given.	1-3
Level 2	Discusses several controls with critique and applies to the scenario.	4-6
Level 3	Discusses several controls with critique, applies to the scenario and references the pre-seen.	7-9

题目 1 (a): 网络风险缓解

级别 1	识别的控制措施很少，几乎未给出理由。	1-3
级别 2	讨论了若干控制措施，并对其进行了评价，适用于该场景。	4-6
级别 3	讨论了若干控制措施，并对其进行了评价，适用于该场景，并引用了预览材料。	7-9

Marking Matrix

Level 1	Few controls identified and few or no reasons given.	1-3
Level 2	Discusses several controls with critique and applies to the scenario.	4-6
Level 3	Discusses several controls with critique, applies to the scenario and references the pre-seen.	7-9

Answer from a student less prepared for the exam!

Student has knowledge but (not working in Rotomyne!)

Senior Finance Manager at Rotomyne who is knowledgeable and competent.

评分矩阵

级别	描述	分数
级别1	识别出的控制措施很少，并且几乎没有或没有给出理由。	1-3分
级别2	讨论了多个控制措施，并进行了评估，应用到具体情景中。	4-6分
级别3	讨论了多个控制措施，并进行了评估，应用到具体情景中并参考了预先给定的信息。	7-9分

级别1：来自没有充分准备考试的学生们的答案！

级别2：学生有知识，但（未在罗托矿业工作！）

级别3：罗托矿业的高级财务经理，知识渊博且能力出众。

Task 1 (b): Mitigation of cyber risks

Protection: User security controls (SCS exam Level 1 answer)

An IT policy document that users must agree to and sign before being given access to the IT systems concerned. This should detail all the regulations for the use of the systems.

User access to IT systems should be identifiable with the use of unique usernames.

Training should still regularly be given to all internal staff to ensure they remain able to detect fraudulent emails, websites and other threats.

Task 1 (b): Mitigation of cyber risks

Protection: User security controls (SCS exam Level 1 answer)

An IT policy document that users must agree to and sign before being given access to the IT systems concerned. This should detail all the regulations for the use of the systems.

User access to IT systems should be identifiable with the use of unique usernames.

Training should still regularly be given to all internal staff to ensure they remain able to detect fraudulent emails, websites and other threats.

第1题 (b): 缓解网络风险

来自准备不充分的学员的回答

- 保护：用户安全控制（SCS考试级别1的答案）

在用户获得相关IT系统访问权限之前，必须同意并签署一份IT政策文件。该文件应详细列出所有使用系统的规章制度。

用户对IT系统的访问应通过唯一的用户名进行识别。

应定期对所有内部员工进行培训，以确保他们能够识别欺诈性邮件、网站和其他威胁。

Task 1 (b): Mitigation of cyber risks

Student has knowledge but
(not working in Rotomyne!)

Protection: User security controls (SCS Level 2 answer – “textbook version”)

An IT policy document that users must agree to and sign before being given access to the IT systems concerned. This should detail all the regulations for the use of the systems. This policy also needs to address external users (e.g. external auditors) who may require access to systems. User access to IT systems should be identifiable with the use of unique usernames.

Users also need to be authenticated, i.e. when a user logs on with their username, they need to prove their identity. This can be done via single-factor authentication, e.g. a strong, unique password, fingerprint or token. To increase security, two-factor authentication can be used, requiring a second identifier in order to gain access.

Training should still regularly be given to all internal staff to ensure they remain able to detect fraudulent emails, websites and other threats. Updates should focus on emerging cybersecurity threats, such as common types of phishing emails being sent around or new forms of social engineering.

第1题 (b): 缓解网络风险

保护：用户安全控制（SCS考试级别2答案 - “教科书版本”）学生有知识，但（未在罗托矿业工作！）

在用户获得相关IT系统访问权限之前，必须同意并签署一份IT政策文件。

该文件应详细列出所有系统使用的规章制度。该政策还需要针对可能需要访问系统的外部用户（如外部审计师）。用户对IT系统的访问应通过唯一的用户名进行识别。

用户还需要进行身份验证，即当用户使用用户名登录时，需要证明其身份。这可以通过单因素身份验证来实现，例如强且唯一的密码、指纹或令牌。为了提高安全性，可以使用双因素身份验证，需要第二个标识符才能获得访问权限。应定期对所有内部员工进行培训，以确保他们能够识别欺诈性邮件、网站和其他威胁。更新内容应关注新兴的网络安全威胁，例如常见的网络钓鱼邮件类型或新形式的社会工程学。

Task 1 (b): Mitigation of cyber risks

Protection: User security controls (SCS exam Level 3 answer **with application skills**)

Rotomyne (or we) should have an IT policy document that users must agree to and sign before being given access to our IT systems. This should detail all the regulations for the use of our systems. This policy also needs to address external users such as our external auditors, who may require access to our systems.

User access to IT systems should be identifiable with the use of unique usernames. Users also need to be authenticated. For instance, when our staff log on with their username, they need to prove their identity. This can be done via single-factor authentication such as a strong/unique password, fingerprint or token. To increase security, we may use two-factor authentication, requiring a second identifier in order to gain access.

While we are a mining and lithium processing organisation, training should still regularly be given to all our internal staff (why?) to ensure they remain able to detect fraudulent emails, websites and other threats. Updates should focus on emerging cybersecurity threats, such as common types of phishing emails being sent around.

第1题 (b): 缓解网络风险

在罗托矿业工作的
资深财务经理，体
现了知识和技能

保护：用户安全控制（SCS考试级别3答案，包含应用技能）罗托矿业的高级财务经理，知识渊博且能力出众。

罗托矿业（或我们）应该有一份IT政策文件，用户在获得访问我们IT系统权限之前，必须同意并签署。这份文件应详细列出我们系统使用的所有规定。该政策还需要针对可能需要访问我们系统的外部用户，例如我们的外部审计师。

IT系统的用户访问应通过唯一的用户名进行识别。用户还需要进行身份验证，即当用户使用用户名登录时，需要证明其身份。这可以通过单因素身份验证实现，例如强且唯一的密码、指纹或令牌。为了提高安全性，可以使用双因素身份验证，要求第二个标识符才能获得访问权限。

虽然我们是一家专注于安全的组织，但仍应定期对所有我们的内部员工进行培训，以确保他们能够识别欺诈性邮件、网站和其他威胁。更新内容应关注新兴的网络安全威胁，例如常见的网络钓鱼邮件类型或新形式的社会工程学。

Task 1 (b): Mitigation of cyber risks

Protection: Data security controls

We need to protect personal, sensitive or valuable data when being stored internally, when they are stored on our servers and transmitted. We need to be confident that we use a very secure encryption method for the data in our cloud servers.

(why?) As remote working becomes a more common working pattern, the risk of vulnerabilities when transmitting data is elevated. When working from home or at our client sites, data is being transferred back and forth between employees and our offices. For this reason, VPNs (virtual private networks) could be securely set up to encrypt Internet traffic, which helps protect data from interception and maintains user anonymity.

In terms of procedures, employees should be locking their screens whenever they are away from their computers, in order to protect data. This is key control, (why?) as many of our geologists and other scientists work remotely at our partner sites, mining sites etc. Given that we have invested in cyber controls, it is likely that these controls may already be in place, but again we must remain vigilant and implement any new controls to address existing or emerging threats.

题目 1 (b): 网络风险缓解

保护：数据安全控制

个人数据、敏感数据或有价值的数据在内部存储、在我们的新云服务器上存储以及传输过程中均需受到保护。对于存储在云服务器上的数据，我们需要确保其使用的是极其安全的加密方法，因为安全性的需求至关重要。

(为什么?) 随着远程工作逐渐成为一种常见的工作模式，数据传输过程中的安全漏洞风险也随之增加。无论是在家工作还是在客户现场，数据都在员工与办公室之间进行传输。因此，建立虚拟私人网络（VPN）以加密互联网流量是一个有效的措施，这不仅可以帮助保护数据不被截获，还能保持用户的匿名性。

就操作程序而言，员工应在离开电脑时锁定屏幕，以防数据泄露。(为什么?) 这一控制措施非常关键，特别是考虑到我们许多风险管理顾问常在客户现场工作。考虑到我们已经在网络安全控制上进行了投资，这些控制措施可能已经实施，但我们仍需保持警觉，不断实施新的控制措施以应对现有的或新出现的威胁。

Task 1 (b): Mitigation of cyber risks

Protection: Software security controls

We should keep the anti-virus software installed on each computer up to date. (why?) New viruses and other malware threats are created daily.

One issue that can lead to increased vulnerability is **our employees** delaying or refusing to install updates to avoid disruption to their work. As such, the IT department should ensure that the computers are configured so that updates happen automatically and in a timely manner.

Where there is outdated software on a computer, IT should be notified and should immediately address the issue before the vulnerability can be exploited by a malicious actor or programme.

Task 1 (b): Mitigation of cyber risks

Protection: Software security controls

We should keep the anti-virus software installed on each computer up to date. (why?) New viruses and other malware threats are created daily.

One issue that can lead to increased vulnerability is **our employees** delaying or refusing to install updates to avoid disruption to their work. As such, the IT department should ensure that the computers are configured so that updates happen automatically and in a timely manner.

Where there is outdated software on a computer, IT should be notified and should immediately address the issue before the vulnerability can be exploited by a malicious actor or programme.

题目 1 (b): 网络风险缓解

保护：软件安全控制

我们应该让每台计算机上安装的防病毒软件保持更新。新的病毒和其他恶意软件威胁每天都在出现。

我们的员工推迟或拒绝安装更新以避免干扰工作，这一行为可能增加安全漏洞。因此，IT部门应确保计算机配置可以自动并及时更新。

如果计算机上的软件过时了，IT部门应被通知并立即解决此问题，防止恶意行为者或程序利用这些漏洞。

Task 1 (b): Mitigation of cyber risks

Protection: Hardware and network security controls

As Rotomyne operates numerous interrelated systems operated by our customers, suppliers, university partners etc., it is important to put firewalls around each system and only allow authorised traffic between systems.

The physical assets of our offices, mines and processing plants need to be kept secure. The physical access system and CCTV coverage in place at the offices will provide security for on-site IT hardware. Strict controls also need to be used when devices are taken off-site. For instance, off-site hardware controls might include a sign-in/sign-out system should be set up for shared devices, and a policy document, which includes responsibility guidelines, signed by employees for all company assets taken off-site.

We must continue to invest to maintain our own physical security because our security specialism should help to protect us against most common forms of physical cybersecurity risk. This is important given the nature of our data and the knowledge of our specialism could attract more advanced threats.

题目 1 (b): 网络风险缓解

保护：硬件和网络安全控制

由于罗托矿业运营着众多由客户、供应商和大学合作伙伴等使用的互联系统，因此为每个系统设置防火墙，并且只允许授权流量在系统之间流通至关重要。

我们的办公室、矿区和加工厂的物理资产必须确保安全。现有的门禁系统和闭路电视监控可以为办公室内的IT硬件提供现场安全保障。对设备进行离场管理时，也应采取严格的控制措施。例如，对于外借设备，可以建立签入/签出系统，并要求员工在带出公司资产时签署包含责任条款的政策文件。

我们必须继续投入资源来维护自身的物理安全，因为我们的安全专长可以帮助我们抵御大多数常见的物理网络安全风险。鉴于我们数据的敏感性以及安全领域的专业性，可能会吸引更高级别的威胁，因此保持高标准的安全措施尤为重要。

Task 1 (b): Mitigation of cyber risks

Detection

If Rotomyne's IT systems come under attack, it is critical there is some method for identifying that this attack is taking place, or has taken place. Our information security risks must be monitored on an ongoing basis, and we must ensure that this includes live monitoring of our networks and systems, with IT experts watching for unusual patterns of activity. (why?) Not every organisation will require this level of vigilance, but the nature of our business operations requires a higher standard of detection.

We should also consider intrusion detection systems to augment our defences against attack. Our cybersecurity risk management must include a policy of detection in place so that if we come under attack, we are aware. In order to do this, the IT department should carry out regular monitoring for unusual patterns of activity. Given the level of sensitivity of our systems to business interruption (especially our supply chain management), due to a cyber attack, we also ensure a suitable intrusion detection and prevention system.

题目 1 (b): 网络风险缓解

检测

如果罗托矿业的IT系统遭到攻击，关键在于能够及时识别攻击正在发生或已经发生。我们的信息安全风险必须得到持续监控，并确保实施网络和系统的实时监控，由IT专家关注任何异常活动模式。（为什么？）并非所有企业都需要如此高的警戒水平，但由于我们的业务性质，要求我们具备更高的检测标准。

我们还应考虑部署入侵检测系统，以增强我们对攻击的防御能力。网络安全风险管理必须包含一套完善的检测机制，确保在遭遇攻击时我们能够及时知晓。为此，IT部门应定期监测异常活动模式。**鉴于我们的系统对业务中断的高度敏感性**（特别是在供应链管理方面），我们还需要确保拥有合适的入侵检测和预防系统。

Task 1 (b): Mitigation of cyber risks

Response

The general consensus today is that a company will suffer a cyber breach at some point. While Rotomyne will take every reasonable measure to prevent this from happening, it's impossible to totally safeguard against these types of risks. Therefore, the IT department needs to prepare itself for this possibility.

Rotomyne should have a Computer Security Incident Response Team (CSIRT) (why?) whose aims are to reduce losses, reduce downtime, support investigations when necessary and facilitate crisis communications. We should also ensure that our contingency management has capability for business continuity and disaster recovery plans so that we can recover in the event of a cyber-attack. It is imperative that we build back trust amongst our clients immediately as assured in our vision statement.

题目 1 (b): 网络风险缓解

响应

现在大家普遍认为公司早晚都会遇到网络安全事件。尽管罗托矿业会尽一切努力来避免这种情况发生，但完全避免风险是不可能的。所以，IT部门要提前做好准备。

罗托矿业应该组建一个计算机安全事件响应团队（CSIRT），(为什么?)它的主要任务是减少损失、缩短停工时间、在需要时帮助调查并协助危机沟通。同时，我们的应急管理体系还需要具备业务连续性和灾难恢复的能力，这样当遭遇网络攻击时，我们可以迅速恢复。最关键的是，我们必须立刻按照愿景声明的承诺，重新赢得客户的信任。

Task 1 (b): Check list

Have we answered the question (**set by the examiner!**)?

”Could you explain how cyber risks can be mitigated?”

Have you related your answer to the pre-seen and unseen material?

Did you think and write as the Senior Finance Manager of Rotomyne?

第1 题(b): 检查清单

我们回答了问题了吗（由考官提出的！）？

“你能解释一下我们可能面临的网络风险类型吗？”

你是否将你的答案与预览材料和未提供的材料联系起来？

你是否以罗托矿业的高级财务经理的身份思考并撰写？

Given the above, I need your input as follows:

- Could you explain the types of cyber risks that we might face, and how they might be mitigated?

[Sub task (1a) - 55%]

- And could you evaluate the different debt finance methods that could be used to fund a digital strategy project, and the impact of the extra debt funding required for the digital strategy project?

45% of 33 marks = 15 marks

So, 7-8 marks for each part of this sub-task. ***[Sub task (1b) - 45%]***

The complete Task and Unseen Material is provided separately.

鉴于此，我需要你就以下几点提供意见：

↵

- 你能解释我们可能面临的网络风险类型以及如何缓解这些风险吗？

↵

【小题 (1a) - 55%】

↵

- 你能评估用于资助数字战略项目的不同债务融资方法，及额外债务融资对该项目的影响吗？

33分的45%等于15分。因此，这个小题的每个部分应该得到7到8分。

【小题 (1b) - 45%】

↵

完整的题目和未事先提供的材料将分开提供。

Task 1 (b): Marking Matrix

Level 1	States few (1 to 2) debt payment methods.	1-3
Level 2	Evaluates several debt payment methods (e.g. loans, revolving credit, debentures), assesses appropriateness to the scenario, and makes basic reference to the pre-seen.	4-6
Level 3	Evaluates several payment methods (e.g. loans, secured loans, revolving credit, debentures, convertible bonds), assesses appropriateness to the scenario, categorises debt methods (bank, capital markets), and references the pre-seen.	7-8

题目 1 (b): 债务融资方法

级别1	列举少量（1到2种）债务支付方法。	1-3
级别2	评估多种债务支付方法（例如贷款、循环信贷、债券），评估其适用性，并基本参考预览材料。	4-6
级别3	评估多种支付方法（例如贷款、有担保贷款、循环信贷、债券、可转换债券），评估其在场景中的适用性，对债务方法进行分类（银行、资本市场），并参考预览材料。	7-8

Task 2 (a): Debt finance methods – Key Points

Evaluation of alternative debt finance methods

Bank loans

Revolving Credit Facilities (RCFs)

Debentures

Convertible debentures

Task 2 (a): Debt finance methods – Key Points

Evaluation of alternative debt finance methods

Bank loans

Revolving Credit Facilities (RCFs)

Debentures

Convertible debentures

题目 2 (a): 债务融资方法 – 要点

评估其他债务融资方法

银行贷款

循环信贷设施（RCFs）

债券

可转换债券

Task 2 (a): Debt finance methods

Bank loans

Bank loans provide a specific amount of funding for a set period of time at either a variable rate (which changes with the market rates) or fixed rate. Loans can be unsecured (for small amounts) or secured (on collateral such as non-current assets) for large amounts. Unsecured loans will be more expensive than secured debt due to the increased risk being taken by the bank. The advantages of bank loans are that they are simple, easy to arrange and have flexible terms and conditions (e.g. repayment of a loan in full at any time with minimal warning).

We already have long term debt of P\$5,000m. As we have adhered to the repayment schedule, our bank may provide additional debt finance. Our current bank loan interest is presently 10% (= P\$500m / P\$5,000m). A disadvantage of bank loans is that the interest charged is linked to regional interest rates (of Prreland), which may be volatile and may adversely change. Another disadvantage is that the bank could also significantly alter the terms of the loan. For example, the bank could demand an earlier repayment of the bank loan provided.

第 2 题(a): 债务融资方法

银行贷款

银行贷款在固定期限内提供一定金额的资金，利率可以是随市场变化的浮动利率或固定利率。贷款分为无抵押贷款（通常金额较小）和有抵押贷款（如以非流动资产作抵押的大额贷款）。由于银行承担的风险较高，无抵押贷款通常比有抵押贷款的利率更高。银行贷款的优势在于其办理简单、容易安排，且条款灵活（例如可以在短期通知的情况下提前全额还款）。

我们目前已有50亿P\$的长期债务，因遵守还款计划，银行可能会提供额外的债务融资。目前贷款的利率为10%（即P\$500m / P\$5,000m）。银行贷款的一个缺点是利率与波兰德的区域利率挂钩，这些利率可能波动较大，产生不利影响。另外，银行也可能会随时调整贷款条款，例如要求提前还款。

Task 2 (a): Debt finance methods

Revolving credit facilities (RCFs)

A revolving credit facility (RCF) is a bank facility that would **allow us to withdraw** funds up to an agreed credit limit. The amount available to use varies as the funds are borrowed or paid (in a similar manner to a credit card).

RCFs are very flexible debt financing options as they allow an entity to minimise interest payments. However, the terms of an RCF (as in a bank loan) are not fixed, so **we could change the credit limit and interest rate at any time.**

第 2 题(a): 债务融资方法

循环信贷额度 (RCFs)

循环信贷额度 (RCF) 是一种银行授信工具，**允许我们**在商定的信用额度内**提取**资金。可用资金会随着借款或还款情况而变化，类似于信用卡的运作方式。

RCF 是一种非常灵活的债务融资选择，因为它可以帮助企业减少利息支出。然而，RCF 的条款（如同银行贷款）并非固定，**因此我们可以随时调整信用额度和利率**。

Task 2 (a): Debt finance methods

Debentures

A debenture represents a medium to long-term borrowing mechanism employed by larger organisations, like ours, to raise funds. This term is often used synonymously with terms such as bond, loan stock or note, indicating that it functions similarly to a loan, confirming the obligation of the company to repay a certain amount with added interest.

Debentures can be secured against the assets of the company. For instance, a fixed charge can be created on Rotomyne's property, plant and machinery; alternatively, a floating charge on trade receivables of Rotomyne.

Typically, debentures can be transferred freely by the holder and might be traded on financial exchanges, such as the Porrrland Stock Market. While debenture holders do not possess voting rights at the company's general meetings, they might participate in separate meetings or votes, for instance, regarding amendments to the rights associated with the debentures.

第 2 题(a): 债务融资方法

债券

债券是一种用于中长期融资的工具，通常被像我们这样的大企业用来筹集资金。这个词常常与“债券”、“贷款股票”或“票据”等同义，表明其与贷款类似，意味着公司有义务偿还一定金额并支付利息。

债券可以通过公司的资产进行担保。例如，可以在罗托矿业的房产、厂房和设备上设立固定抵押，或者在应收账款上设立浮动抵押。债券通常可以由持有人自由转让，并且可以在金融交易所（如波兰德证券交易所）交易。虽然债券持有人没有公司股东大会的投票权，但他们可以在一些独立会议中参与投票，比如关于修改债券相关权利的决议。

Task 2 (a): Debt finance methods

Convertible Debentures

Convertible debentures are debentures which can be converted into **our equity shares** after a predetermined period of time. 'Convertibility' is a feature that **we may add to the bonds we would issue to make them more attractive to our potential investors** as the debt holder has the opportunity to make a profit if our share price rises significantly.

Due to the conversion feature's benefit to investors, convertible bonds often carry lower interest rates compared to non-convertible corporate bonds. **Moreover, if we issue convertible bonds (as opposed to traditional debt mechanisms, such as bonds or bank loans), they may leverage Rotomyne's accounting ratios and lack voting rights.**

第 2 题(a): 债务融资方法

可转换债券

可转换债券是指可以在特定时间后转换为**我们公司股票**的债券。“可转换性”是**我们可以为发行债券添加的一个特点，以吸引潜在的投资者**，因为如果我们的股价大幅上涨，债权人就有机会获利。

由于这一转换特性给投资者带来了好处，可转换债券的利率通常比不可转换的公司债券要低。此外，若我们发行可转换债券（而不是传统的债务工具，如债券或银行贷款），这可能会优化罗托矿业的财务比率，同时这些债券不具有投票权。

Task 1 (b): Check list

Have we answered the question (**set by the examiner!**)?

”Could you evaluate the debt financing methods?”

Have you related your answer to the pre-seen and unseen material?

Did you think and write as the Senior Finance Manager of Rotomyne?

第1 题(b): 检查清单

我们回答了问题了吗（由考官提出的！）？

“你能评估一下各种债务融资方式吗？”

你是否将你的答案与预览材料和未提供的材料联系起来？

你是否以罗托矿业的高级财务经理的身份思考并撰写？

Task 1 (b): Marking Matrix

Level 1	States few (1 to 2) debt impact areas.	1-3
Level 2	Discusses several debt impact areas (gearing, financial risk, capital structure, security), references figures, and relates to the scenario.	4-5
Level 3	Discusses several debt impact areas (gearing, financial risk, capital structure, security), references figures, relates to scenario, and references pre-seen.	6-7

题目 1 (b):评分矩阵

级别1	列举少量（1到2个）债务影响领域。	1-3
级别2	讨论多个债务影响领域（财务杠杆、财务风险、资本结构、担保），引用数据，并与情景相关。	4-5
级别3	讨论多个债务影响领域（财务杠杆、财务风险、资本结构、担保），引用数据，与情景相关，并引用预览材料。	6-7

Task 2 (a): Impact of extra debt funding – Key Points

Evaluation of Impact of extra debt funding

gearing - present position

Gearing after the loan

Target capital structure – WACC – M&M

Security – should we give and what we may give?

第2 题(a): 额外债务融资的影响 — 关键点

评估 额外债务融资影响

当前的负债率状况

贷款后的负债率

目标资本结构 – 加权平均资本成本（WACC）– M&M理论

担保 – 我们是否需要提供担保，若需要，我们可以提供什么类型的担保？

Task 2 (a): Impact of extra debt funding

Evaluation of Impact of extra debt funding

(point #1) Gearing and financial risk

Gearing will be increased by debt finance and lowered with equity, although the latter is a more onerous process. Financial risk will increase with gearing, (why?) because the interest charge is payable whether or not Rotomyne is making a profit, unlike the payment of dividends. However, debt will improve returns as the interest charge is tax deductible and is also generally lower than the cost of equity.

We will need to assess the level of financial risk Rotomyne is willing to take from debt, via a shareholder and operational perspective. The riskier we are willing to be, the more debt we can introduce and vice versa. It is influenced by our risk appetite.

第2 题(a): 额外债务融资的影响

评估额外债务融资影响

(要点#1) 负债率与财务风险

通过债务融资可以提升负债率，而股权融资则会降低负债率，虽然股权融资过程更加复杂。负债率的增加会带来更高的财务风险，（为什么？）这是因为，无论罗托矿业是否盈利，都必须支付利息费用，和股息支付方式不同。然而，债务可以提高回报，因为利息支出可以抵扣税费，而且债务融资的成本通常低于股权融资。

我们需要从股东和运营的角度来评估罗托矿业愿意承担的财务风险。我们愿意承担的风险越大，能增加的债务就越多，反之亦然。这将受我们风险偏好的影响。

Task 2 (a): Impact of extra debt funding

Evaluation of Impact of extra debt funding

(point #2) Gearing before and after the loan

At present levels, Rotomyne has a gearing of 104% (= LT debt/equity) With an extra P\$100m of debt finance, the gearing will increase to 106%.

This marginally increases our financial risk but way above the gearing of our global competitor, Lithidig which has a current gearing of 34%.

The additional P\$100m of debt may also marginally increase our cost of finance. Our current cost of finance is 10%, however we may be able to reduce this by offering debt collateral for the additional loan and debt restructuring.

第2题 (a): 额外债务融资的影响

评估额外债务融资影响

(要点#2) 贷款前后的负债率

目前，罗托矿业的负债率为104%（长期债务/股本比）。如果我们再新增1亿B\$的债务融资，负债率将上升至106%。

虽然这只会轻微增加我们的财务风险，但依然远高于我们全球竞争对手利恩迪格的34%负债率。

此外，新增的1亿B\$债务可能也会略微增加我们的融资成本。当前我们的融资成本是10%，不过如果我们能为新增的贷款提供抵押，并进行债务重组，融资成本可能会有所降低。

Task 2 (a): Impact of extra debt funding

Evaluation of Impact of extra debt funding

(point #3) Target capital structure

The average cost of the capital for a business is worked out based on the cost of the capital and its proportion (weight) and is referred to as the weighted average cost of capital (WACC). It is essentially the average return we should make from any investment, to satisfy all shareholders and debt holders. Based on Modigliani and Miller theory, the WACC is lowest at an optimal level, and also maximises the value of the firm.

At Rotomyne, we should seek to minimise our WACC and hence maximise our shareholder returns. Thus, a proportion of debt will improve the WACC as debt is cheaper than equity.

第2 题(a): 额外债务融资的影响

评估额外债务融资影响

(要点#3) 目标资本结构

企业的平均资本成本是通过资本成本及其权重比例计算得出的，这就是所谓的加权平均资本成本（WACC）。本质上，它是在任何投资中应实现的平均回报，以满足股东和债权人的要求。根据莫迪利亚尼和米勒的理论，WACC在最优水平时最低，并且可以最大化企业价值。

在罗托矿业，我们应该力求降低WACC，从而最大化股东回报。因此，适量的债务会有助于降低WACC，因为债务成本通常低于股本成本。

Task 2 (a): Impact of extra debt funding

Evaluation of Impact of extra debt funding

(point #4) Availability of security

If we were to finance the investment via additional debt funding, this may have to be secured against our assets. At present, we have P\$7,270m in the form of property, plant and equipment (PPE). This appears to offer a good degree of security.

However, the existing debt of P\$5,000m would most likely have been secured on our assets (a floating charge or a fixed charge on specific assets). So, I need to investigate on the remaining assets (within our PPE) that can act as security for the debt funding. That said, it seems unlikely that our bank has not protected itself with a restrictive covenant limiting our ability to borrow further and to pledge assets as security. Given that the investment amount is relatively small, there may be some flexibility with the bank.

第 2 题(a): 额外债务融资的影响

评估额外债务融资影响

(要点 #4) 担保的可行性

如果我们通过额外的债务融资来进行这项投资，可能需要用我们的资产作为担保。目前，我们拥有72.7亿B\$的物业、厂房和设备（PPE），这为提供担保看起来具备一定的优势。

不过，我们现有的50亿B\$债务很可能已经以这些资产作为担保，可能是浮动抵押或特定资产上的固定抵押。因此，我需要进一步调查PPE资产中是否还有剩余部分可以用来作为新债务的担保。不过，银行很可能已经通过限制性条款保护了自身权益，限制了我们进一步借款和抵押资产的能力。考虑到投资金额相对较小，或许在与银行协商时能够获得一些灵活性。

Task 1 (b): Check list

Have we answered the question (**set by the examiner!**)?

”Could you evaluate the impact of the extra P\$ 100 million debt funding?”

Have you related your answer to the pre-seen and unseen material?

Did you think and write as the Senior Finance Manager of Rotomyne?

Task 1 (b): Check list

Have we answered the question (**set by the examiner!**)?

”Could you evaluate the impact of the extra P\$ 100 million debt funding?”

Have you related your answer to the pre-seen and unseen material?

Did you think and write as the Senior Finance Manager of Rotomyne?

第1 题(b): 检查清单

我们回答了问题了吗（由考官提出的！）？

“你能评估一下新增1亿B\$债务融资的影响吗？”

你的回答是否结合了案例预览材料和未公开的信息？

你是否以罗托矿业的高级财务经理的身份思考并撰写？